

Physical Chemistry (II)

Lecture 12

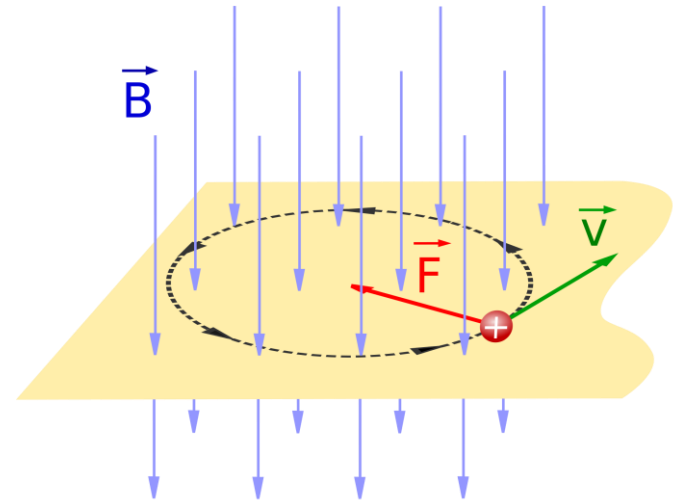
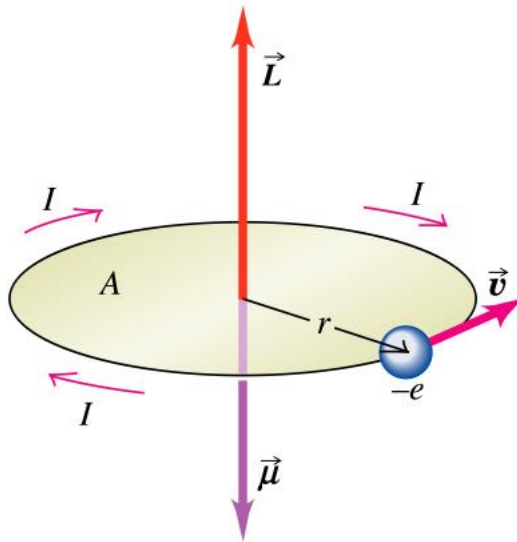
Zeeman Effect

Perturbation Theory



Last class

Electron as a Magnet



$$\vec{\mu} = -\frac{e}{2}(\vec{r} \times \vec{v}) = -\frac{e}{2m}\vec{L}$$

$$V = \frac{eB}{2m}L_z$$

Quantum Mechanics

$$\left(-\frac{\hbar^2}{2\mu} \nabla^2 + \mathcal{V} \right) \psi = E\psi$$

If a hydrogen-atom electron is in a magnetic field,

$$\left(\underbrace{-\frac{\hbar^2}{2\mu} \nabla^2 - \frac{1}{4\pi\epsilon_0} \frac{Ze^2}{r}}_{\text{Hamiltonian for H atom}} + \underbrace{\frac{eB}{2\mu} \hat{L}_z}_{\text{Additional Hamiltonian}} \right) \psi = E\psi$$

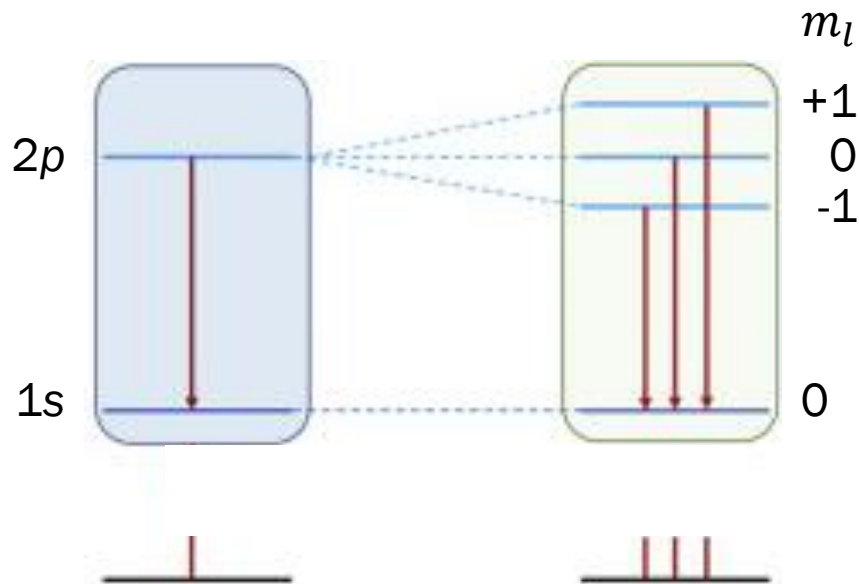
Hamiltonian for H atom
 $\hat{H}_0$

Additional Hamiltonian
 $\hat{H}_{\text{magnet}}$

The eigenfunctions for $\hat{H}_0$ are also those for $\hat{H}_{\text{magnet}}$

Effect of Magnetic Field

$$E = -\frac{ZR_H}{n^2} + \beta_B m_l B$$

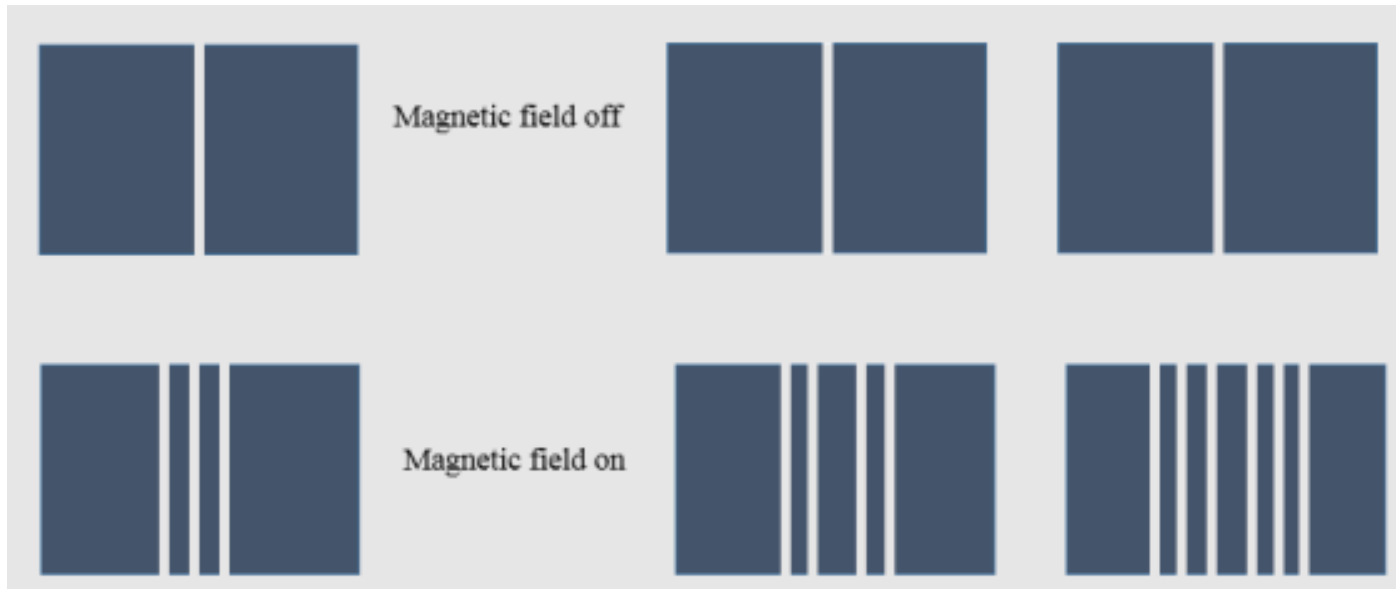


Last class

Zeeman Effect

Normal Zeeman effect

Anomalous Zeeman effect



“Spin” Angular Momentum

An electron has *intrinsic angular momentum*

= an electron intrinsically behaves like a tiny magnet

- Eigenfunction $|s m_s\rangle$

$$\hat{S}^2 |s m_s\rangle = s(s + 1)\hbar^2 |s m_s\rangle$$

$$\hat{S}_z |s m_s\rangle = m_s \hbar |s m_s\rangle$$

- Quantum numbers s, m_s

s : fixed (intrinsic)

$$m_s = -s, -(s - 1), \dots, (s - 1),$$

Last class

Free Electron in Magnetic Field

- For an electron, $s = 1/2$

$$m_s = \pm 1/2$$

- Orbital angular momentum energy

$$E_{\text{orbital}} = \beta_B m_l B$$

- Spin angular momentum energy

$$E_{\text{spin}} = g\beta_B m_s B$$

- $g \approx 2$ (g-factor): experimentally required



Last class

Full Description of H-Atom Electron

- Quantum numbers: n, l, m_l, m_s (s is fixed)
- Wavefunction:

$$\psi_{nlm_l}(r, \theta, \phi) |m_s\rangle = |n \ l \ m_l \ m_s\rangle$$

- Energy (no magnetic field):

$$E_n = -\frac{ZR_H}{n^2}$$

- Energy (in a magnetic field):

$$E = -\frac{ZR_H}{n^2} + ?$$



Last class

In a Strong Field

Two magnetic moments independently react to the field

$$E_{\text{magnet}} = E_{\text{orbital}} + E_{\text{spin}}$$

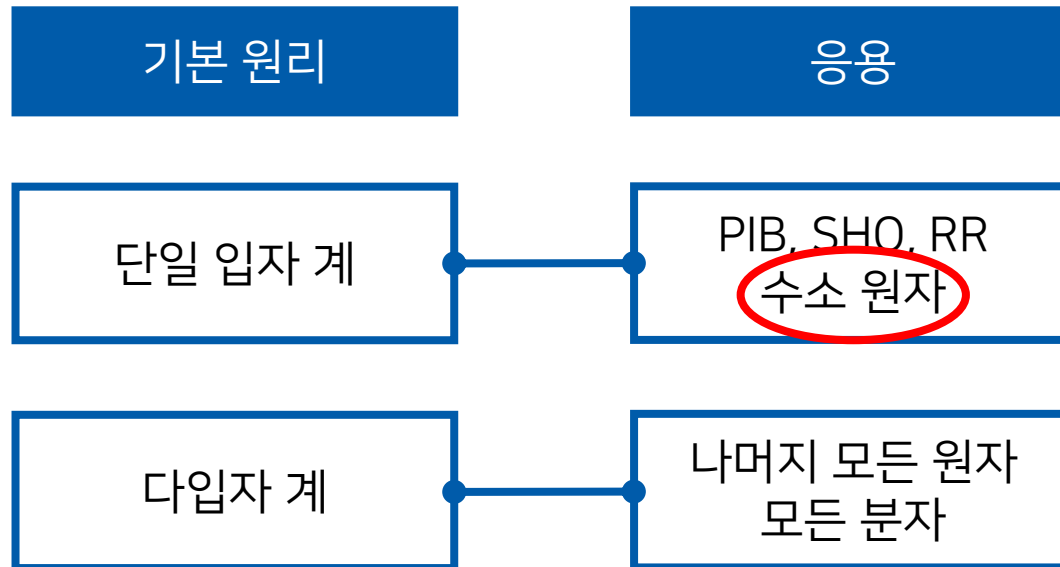
$$E_{\text{orbital}} = \beta_B m_l B$$

$$E_{\text{spin}} = g\beta_B m_s B$$

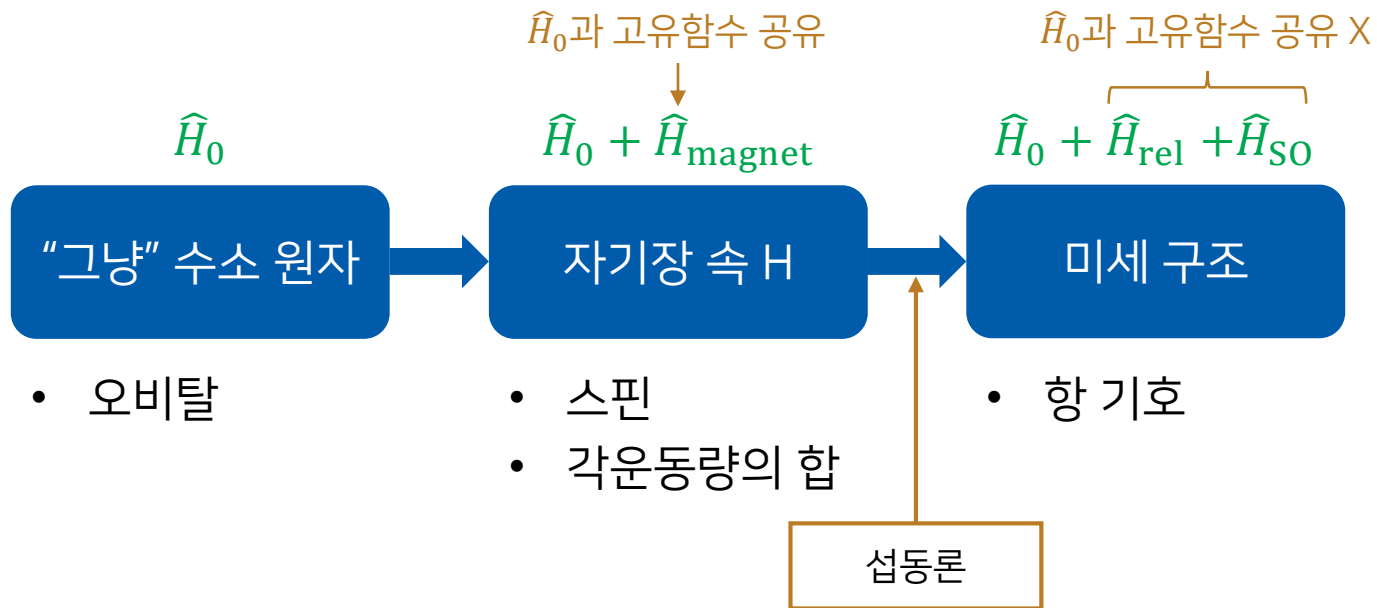
$$E = E_n + E_{\text{magnet}} = -Z \frac{R_H}{n^2} + \beta_B B (m_l + 2m_s)$$



Where Are We?



Where Are We: Hydrogen Atom

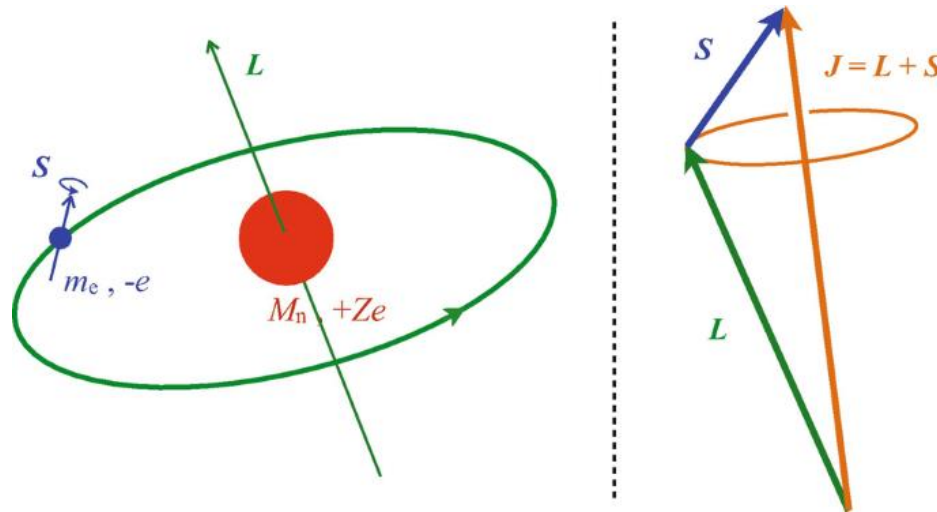


M 7.6

* Griffiths 7.4

Total Angular Momentum

- Orbital and spin angular momenta can be added
- Angular momenta are vectors ($\vec{L}$ and $\vec{S}$)
- The total angular momentum $\vec{J} = \vec{L} + \vec{S}$ is conserved



(Image: Kastberg, "The Spin-Orbit Interaction", in *Structure of Multielectron Atoms*, 2020)

QM of Total Angular Momentum

- Eigenfunctions $|j\ m_j\rangle$
- Eigenvalues

$$\hat{J}^2 |j\ m_j\rangle = j(j+1)\hbar^2 |j\ m_j\rangle$$

$$\hat{J}_z |j\ m_j\rangle = m_j\hbar |j\ m_j\rangle$$

- Quantum numbers
 - Allowed $m_j = -j, -(j-1), \dots, (j-1), j$
 - Allowed $j = (l+s), (l+s-1), \dots, |l-s|$

Examples

Allowed $j = (l + s), (l + s - 1), \dots, |l - s|$

- $l = 0$ (s orbitals): $j = 1/2$
 - $j = 1/2 \rightarrow m_j = \pm 1/2$
- $l = 1$ (p orbitals): $j = 3/2, 1/2$
 - $j = 3/2 \rightarrow m_j = \pm 3/2, \pm 1/2$
 - $j = 1/2 \rightarrow m_j = \pm 1/2$
- $l = 2$ (d orbitals): $j = 5/2, 3/2$
 - $j = 5/2 \rightarrow m_j = \pm 5/2, \pm 3/2, \pm 1/2$
 - $j = 3/2 \rightarrow m_j = \pm 3/2, \pm 1/2$

Two Descriptions of States

	m_l and m_s	j and m_j
1s ($l = 0$)	(0, +1/2), (0, -1/2)	(1/2, +1/2), (1/2, -1/2)
2s ($l = 0$)	(0, +1/2), (0, -1/2)	(1/2, +1/2), (1/2, -1/2)
2p ($l = 1$)	(+1, +1/2), (+1, -1/2) (0, +1/2), (0, -1/2) (-1, +1/2), (-1, -1/2)	(3/2, +3/2), (3/2, -3/2) (3/2, +1/2), (3/2, -1/2) (1/2, +1/2), (1/2, -1/2)
3s ($l = 0$)	(0, +1/2), (0, -1/2)	(1/2, +1/2), (1/2, -1/2)
3p ($l = 1$)	(+1, +1/2), (+1, -1/2) (0, +1/2), (0, -1/2) (-1, +1/2), (-1, -1/2)	(3/2, +3/2), (3/2, -3/2) (3/2, +1/2), (3/2, -1/2) (1/2, +1/2), (1/2, -1/2)
3d ($l = 2$)	10 states	10 states

Orbital + Spin under $\vec{B}$

$$\vec{\mu}_{\text{orbit}} = -\frac{e}{2m}\vec{L}, \quad \vec{\mu}_{\text{spin}} = -\frac{eg}{2m}\vec{S} \approx -\frac{e}{m}\vec{S}$$

From the potential energy $V = -\vec{\mu} \cdot \vec{B}$,

$$V = -\left(-\frac{e}{2m}\vec{L} - \frac{e}{m}\vec{S}\right) \cdot \vec{B} = \frac{e}{2m}(\vec{L} + 2\vec{S}) \cdot \vec{B}$$

$$\hat{H} = -\frac{\hbar^2}{2\mu}\nabla^2 - \frac{1}{4\pi\epsilon_0}\frac{Ze^2}{r} + \frac{e}{2\mu}(\vec{L} + 2\vec{S}) \cdot \vec{B}$$

external field + induced field

Not in text

Weak-Field Hamiltonian

$$\hat{H} = -\frac{\hbar^2}{2\mu} \nabla^2 - \frac{1}{4\pi\epsilon_0} \frac{Ze^2}{r} + \frac{e}{2\mu} (\vec{L} + 2\vec{S}) \cdot \vec{B}$$

Here, $\vec{L} + 2\vec{S} = \vec{J} + \vec{S}$ and on average,

$$\vec{S}_{\text{avg}} = \vec{S} \cdot \left(\frac{\vec{J}}{|\vec{J}|} \right) \frac{\vec{J}}{|\vec{J}|}$$

$$\vec{L} + 2\vec{S} \approx \vec{J} + \vec{S}_{\text{avg}} = \vec{J} \left(1 + \frac{\vec{S} \cdot \vec{J}}{J^2} \right)$$



Not in text

Vector Calculation

$$\vec{L} + 2\vec{S} \approx \vec{J} + \vec{S}_{\text{avg}} = \vec{J} \left(1 + \frac{\vec{S} \cdot \vec{J}}{J^2} \right)$$

Since $\vec{L} = \vec{J} - \vec{S}$, $L^2 = J^2 + S^2 - 2\vec{S} \cdot \vec{J}$ and

$$\vec{S} \cdot \vec{J} = \frac{1}{2} (J^2 + S^2 - L^2)$$

Now,

$$\vec{L} + 2\vec{S} \approx \vec{J} \left[1 + \frac{1}{2J^2} (J^2 + S^2 - L^2) \right]$$

Weak-Field Hamiltonian

$$\hat{H} = -\frac{\hbar^2}{2\mu} \nabla^2 - \frac{1}{4\pi\epsilon_0} \frac{Ze^2}{r} + \frac{e}{2\mu} \left[1 + \frac{1}{2\hat{j}^2} (\hat{j}^2 + \hat{s}^2 - \hat{l}^2) \right] \vec{j} \cdot \vec{B}$$

If we choose the z-axis to lie along $\vec{B}$,

$$\hat{H} = -\frac{\hbar^2}{2\mu} \nabla^2 - \frac{1}{4\pi\epsilon_0} \frac{Ze^2}{r} + \frac{eB}{2\mu} \left[1 + \frac{1}{2\hat{j}^2} (\hat{j}^2 + \hat{s}^2 - \hat{l}^2) \right] \hat{j}_z$$

$$E = -Z \frac{R_H}{n^2} + \beta_B B \left(1 + \frac{j(j+1) + s(s+1) - l(l+1)}{2j(j+1)} \right) m_j$$

Landé g factor $g(j, l)$

g Factor Calculation

$$g(j, l) = 1 + \frac{j(j+1) + s(s+1) - l(l+1)}{2j(j+1)}$$

For $l = 1$,

- $j = 3/2$:

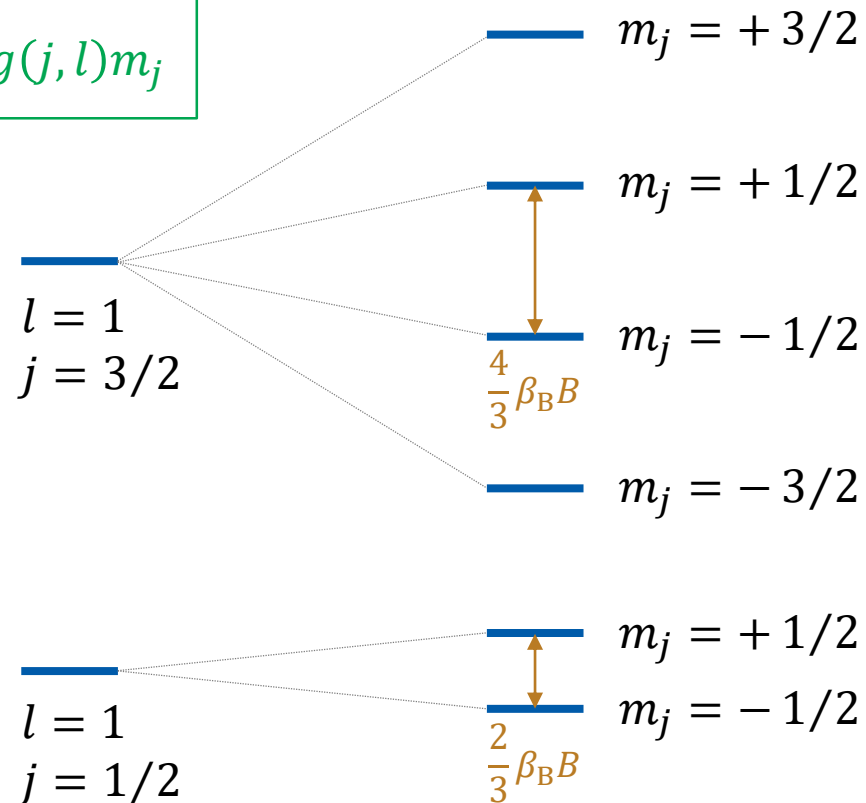
$$g(j, l) = 1 + \frac{(3/2)(3/2+1) + (1/2)(1/2+1) - 1(1+1)}{2 \cdot (3/2)(3/2+1)} = 1 + \frac{1}{3} = \frac{4}{3}$$

- $j = 1/2$:

$$g(j, l) = 1 + \frac{(1/2)(1/2+1) + (1/2)(1/2+1) - 1(1+1)}{2 \cdot (1/2)(1/2+1)} = 1 - \frac{1}{3} = \frac{2}{3}$$

Weak-Field Energy Levels

$$E = -Z \frac{R_H}{n^2} + \beta_B B g(j, l) m_j$$



Field off

Field on

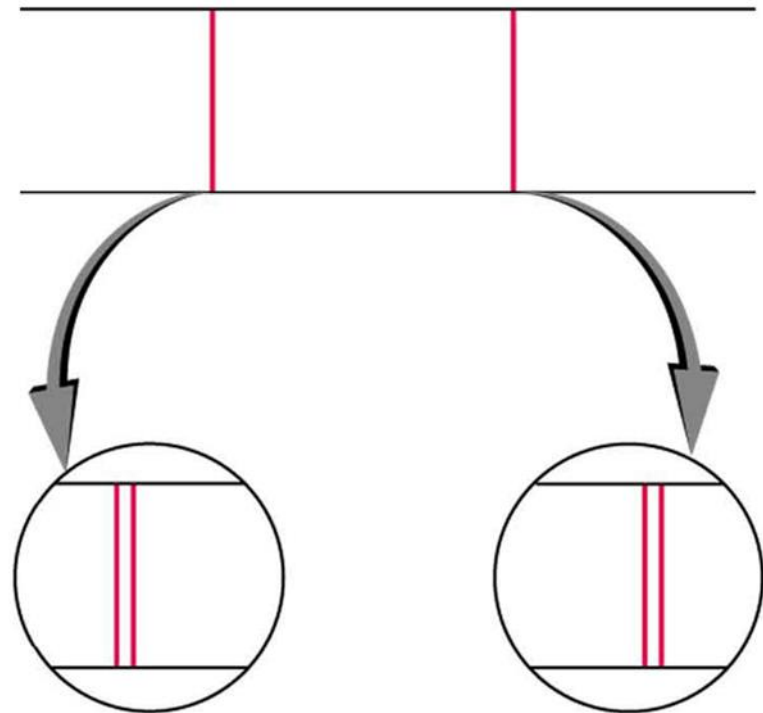
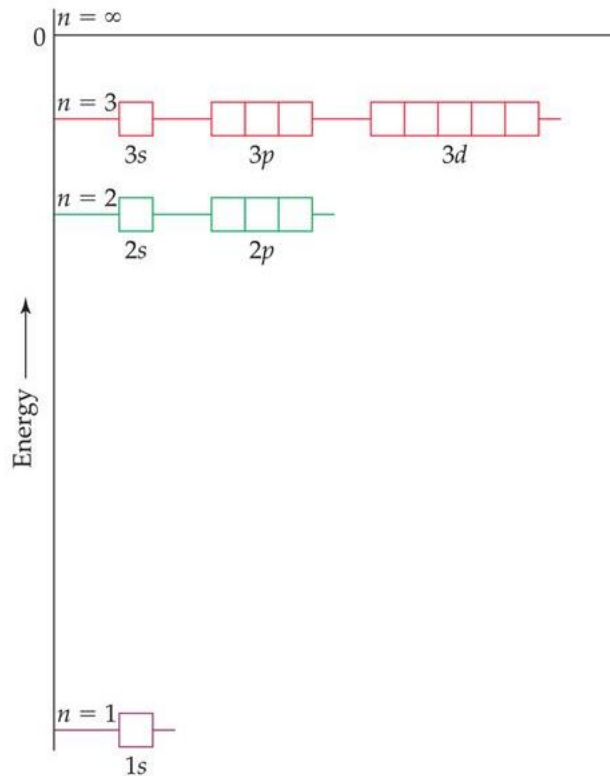
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"Good" Quantum Numbers

Two sets of quantum numbers: (n, l, m_l, m_s) and (n, l, j, m_j)

- Strong $\vec{B}$: (n, l, m_l, m_s) describes the state better
- Weak $\vec{B}$: (n, l, j, m_j) describes the state better

Fine Structure of Hydrogen



Origin of Fine Structure

1. Relativistic effect
2. Spin-orbit coupling

$$\hat{H}_0 \rightarrow \hat{H}_0 + \hat{H}_{\text{rel}} + \hat{H}_{\text{so}}$$

$\hat{H}_{\text{rel}}$ or $\hat{H}_{\text{so}}$ do NOT share eigenfunctions with $\hat{H}_0$!

(Note: the two effects are relatively small)

Relativistic Effect

- Kinetic energy in CM

$$T = \frac{p^2}{2m}$$

- Kinetic energy in relativity

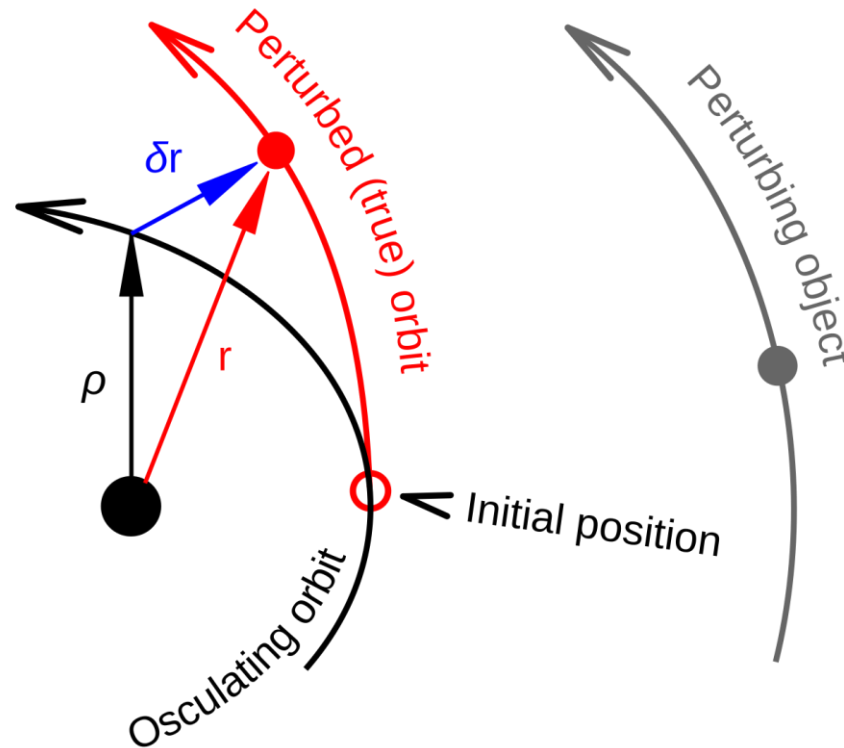
$$T = \sqrt{p^2 c^2 + m^2 c^4} - mc^2 = \frac{p^2}{2m} \overset{\hat{H}_{\text{rel}}}{\text{---} \frac{p^4}{8m^3 c^2} + \dots}$$

Our $|n \ l \ m_l \ m_s\rangle$ is not an eigenfunction of $\hat{p}^4$

How can we treat this?

RS 5.4
M 8.5
* Griffiths 7.1

“Perturbation”



(Image: https://commons.wikimedia.org/wiki/File:Enckes_method-vector.svg)

Perturbation Theory

$$\hat{H}\psi = E\psi$$

We can apply the perturbation theory if the Hamiltonian can be decomposed as

$$\hat{H} = \overset{\substack{\text{unperturbed Hamiltonian} \\ \downarrow}}{\hat{H}_0} + \overset{\substack{\uparrow \\ \text{perturbation}}}{\hat{H}'}$$

where we know $E_n^{(0)}$ and $\psi_n^{(0)}$ (n : quantum number):

$$\hat{H}_0\psi_n^{(0)} = E_n^{(0)}\psi_n^{(0)}$$

- Note: $\hat{H}_0$ and $\hat{H}'$ do not share the eigenfunctions

Series Expansion

Introduce the bookkeeping parameter, λ :

$$\hat{H} = \hat{H}_0 + \lambda \hat{H}'$$

and assume that

$$\psi_n = \psi_n^{(0)} + \lambda \psi_n^{(1)} + \lambda^2 \psi_n^{(2)} + \dots$$

$$E_n = E_n^{(0)} + \lambda E_n^{(1)} + \lambda^2 E_n^{(2)} + \dots$$

Now substitute everything into

$$\hat{H}\psi_n = E_n\psi_n$$

Collecting Terms

$$\begin{aligned} & (\hat{H}_0 + \lambda \hat{H}') \left[\psi_n^{(0)} + \lambda \psi_n^{(1)} + \lambda^2 \psi_n^{(2)} + \dots \right] \\ &= \left[E_n^{(0)} + \lambda E_n^{(1)} + \lambda^2 E_n^{(2)} + \dots \right] \left[\psi_n^{(0)} + \lambda \psi_n^{(1)} + \lambda^2 \psi_n^{(2)} + \dots \right] \end{aligned}$$

Collecting like powers of λ ,

$$\begin{aligned} & \hat{H}_0 \psi_n^{(0)} + \lambda \left[\hat{H}_0 \psi_n^{(1)} + \hat{H}' \psi_n^{(0)} \right] + \lambda^2 \left[\hat{H}_0 \psi_n^{(2)} + \hat{H}' \psi_n^{(1)} \right] + \dots \\ &= E_n^{(0)} \psi_n^{(0)} + \lambda \left[E_n^{(0)} \psi_n^{(1)} + E_n^{(1)} \psi_n^{(0)} \right] \\ & \quad + \lambda^2 \left[E_n^{(0)} \psi_n^{(2)} + E_n^{(1)} \psi_n^{(1)} + E_n^{(2)} \psi_n^{(0)} \right] + \dots \end{aligned}$$

Different Orders

Zeroth order (λ^0):

$$\hat{H}_0 \psi_n^{(0)} = E_n^{(0)} \psi_n^{(0)} \text{ (what we knew)}$$

First order (λ^1):

$$\hat{H}_0 \psi_n^{(1)} + \hat{H}' \psi_n^{(0)} = E_n^{(0)} \psi_n^{(1)} + E_n^{(1)} \psi_n^{(0)}$$

Second order (λ^2):

$$\hat{H}_0 \psi_n^{(2)} + \hat{H}' \psi_n^{(1)} = E_n^{(0)} \psi_n^{(2)} + E_n^{(1)} \psi_n^{(1)} + E_n^{(2)} \psi_n^{(0)}$$

$\vdots$

First-Order Correction

$$\hat{H}_0 \psi_n^{(1)} + \hat{H}' \psi_n^{(0)} = E_n^{(0)} \psi_n^{(1)} + E_n^{(1)} \psi_n^{(0)}$$

Multiply both sides by $\psi_n^{(0)*}$ and integrate;

$$\begin{aligned} \int \psi_n^{(0)*} \hat{H}_0 \psi_n^{(1)} d\tau + \int \psi_n^{(0)*} \hat{H}' \psi_n^{(0)} d\tau \\ = E_n^{(0)} \int \psi_n^{(0)*} \psi_n^{(1)} d\tau + E_n^{(1)} \underbrace{\int \psi_n^{(0)*} \psi_n^{(0)} d\tau}_{= 1 \text{ (normalized)}} \end{aligned}$$

Some Math

$$\int \psi_n^{(0)*} \hat{H}_0 \psi_n^{(1)} d\tau + \int \psi_n^{(0)*} \hat{H}' \psi_n^{(0)} d\tau = E_n^{(0)} \int \psi_n^{(0)*} \psi_n^{(1)} d\tau + E_n^{(1)}$$

For a QM (Hermitian) operator $\hat{A}$, we have

$$\int f^* (\hat{A}g) d\tau = \int (\hat{A}f)^* g d\tau$$

Hence, the first term in LHS becomes

$$\int \psi_n^{(0)*} \hat{H}_0 \psi_n^{(1)} d\tau = \int [\hat{H}_0 \psi_n^{(0)}]^* \psi_n^{(1)} d\tau = E_n^{(0)*} \int \psi_n^{(0)*} \psi_n^{(1)} d\tau$$

which cancels the first term in RHS.

Finally

$$\cancel{\int \psi_n^{(0)*} \hat{H}_0 \psi_n^{(1)} d\tau} + \int \psi_n^{(0)*} \hat{H}' \psi_n^{(0)} d\tau = \cancel{E_n^{(0)} \int \psi_n^{(0)*} \psi_n^{(1)} d\tau} + E_n^{(1)}$$

Therefore,

$$E_n^{(1)} = \int \psi_n^{(0)*} \hat{H}' \psi_n^{(0)} d\tau$$

One can also obtain (just for your information)

$$\psi_n^{(1)} = \sum_{m \neq n} \frac{\int \psi_m^{(0)*} \hat{H}' \psi_n^{(0)} d\tau}{E_n^{(0)} - E_m^{(0)}} \psi_m^{(0)}$$

Example 8-5

Calculate the first-order correction to the ground-state energy of an anharmonic oscillator whose potential is

$$V(x) = \frac{1}{2}kx^2 + \frac{1}{6}\gamma_3x^3 + \frac{1}{24}\gamma_4x^4.$$

The perturbation is

$$\hat{H}' = \frac{1}{6}\gamma_3x^3 + \frac{1}{24}\gamma_4x^4$$

and the ground-state wavefunction of a SHO is

$$\psi_0(x) = (\alpha/\pi)^{1/4} e^{-\alpha x^2/2} \text{ (where } \alpha = (k\mu)^{1/2}/\hbar)$$

Hence,

$$E_0^{(1)} = \int_{-\infty}^{\infty} \psi_0^{(0)*} \hat{H}' \psi_0^{(0)} dx = \left(\frac{\alpha}{\pi}\right)^{1/2} \int_{-\infty}^{\infty} e^{-\alpha x^2/2} \left(\frac{1}{6}\gamma_3x^3 + \frac{1}{24}\gamma_4x^4\right) e^{-\alpha x^2/2} dx$$

Example 8-5

Calculate the first-order correction to the ground-state energy of an anharmonic oscillator whose potential is

$$V(x) = \frac{1}{2}kx^2 + \frac{1}{6}\gamma_3x^3 + \frac{1}{24}\gamma_4x^4.$$

$$E_0^{(1)} = \left(\frac{\alpha}{\pi}\right)^{1/2} \left(\frac{\gamma_3}{6} \int_{-\infty}^{\infty} x^3 e^{-\alpha x^2} dx + \frac{\gamma_4}{24} \int_{-\infty}^{\infty} x^4 e^{-\alpha x^2} dx \right)$$

The first term is zero, as the integrand is an odd function. For the second term, we use the table of integrals to obtain

$$E_0^{(1)} = \left(\frac{\alpha}{\pi}\right)^{1/2} \times \frac{\gamma_4}{24} \times \frac{3\pi^{1/2}}{4\alpha^{5/2}} = \frac{\gamma_4}{32\alpha^2}$$

Where Are We: Hydrogen Atom

